

In particular, we have for the empirical measure $\rho_m = \frac{1}{m} \sum_{i=1}^m \delta_{\text{PD}(\mathcal{X}_i)}$ based on some $\mathbb{R}^d$ -valued point cloud process $\mathcal{X} = (\mathcal{X}_t)_t$

$$F_{\rho_m}(V) = \frac{1}{m} \sum_{i=1}^m W_r^2(\text{PD}(\mathcal{X}_i), V), \quad V \in \mathcal{D}_r.$$

Assuming stationarity of an underlying $\mathbb{R}^d$ -valued point cloud process $\mathcal{X} = (\mathcal{X}_t)_t$ implies that the sequence $(\text{PD}(\mathcal{X}_t))_t$ of corresponding persistence diagrams of a given feature dimension $k \in \{0, \dots, d-1\}$ is also a stationary process in the metric space $(\mathcal{D}_r, W_r)$, that is $\text{PD}(\mathcal{X}_t) \sim \rho_{\mathcal{X}}$ for some probability measure $\rho_{\mathcal{X}}$. Given a sample $\mathcal{X}_1, \dots, \mathcal{X}_m$ stemming from the process $\mathcal{X}$, this suggests

$$\arg \min_{V \in \mathcal{D}_r} \frac{1}{m} \sum_{i=1}^m W_r^2(\text{PD}(\mathcal{X}_i), V) \text{ and } \widehat{\text{Var}}(\mathcal{X}) = \frac{1}{m} \sum_{i=1}^m W_r^2(\text{PD}(\mathcal{X}_i), \widehat{\mu}_{\mathcal{X}})$$

as suitable estimates for the Fréchet mean $\mu(\mathcal{X}) = \mu_{\rho_{\mathcal{X}}}$ and the Fréchet variance $\text{Var}(\mathcal{X}) = \text{Var}_{\rho_{\mathcal{X}}}$ of the persistence diagram $\text{PD}(\mathcal{X}_0)$.

An alternative measure for the dispersion of persistence diagrams corresponding to a (stationary) point cloud process $\mathcal{X} = (\mathcal{X}_t)_t$ is given by the so-called *independent copy (inco)-variance*

$$\sigma^2(\mathcal{X}) := \frac{1}{2} \mathbb{E}[W_r^2(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}'))], \quad (1.4)$$

where $\mathcal{X}'$ is independent of $\mathcal{X}$ and has the same distribution as $\mathcal{X}_0$. The motivation for (1.4) stems from the simple variance identity $\text{Var}(X) = \mathbb{E}[(X - X')^2]/2$, where X is an $\mathbb{R}$ -valued random variable with finite second moment and X' is an independent copy. However, contrary to the Euclidean case, the Fréchet variance and inco-variance do not coincide in general, but a simple calculation shows

$$\text{Var}(\mathcal{X}) - \mathbb{E}[W_r(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}'))]^2 \leq \sigma^2(\mathcal{X}) \leq 2 \text{Var}(\mathcal{X}).$$

Since determining $\text{Var}(\mathcal{X})$ requires a computation or estimation of $\mu(\mathcal{X})$, which in turn leads to a complex minimization problem with possibly non-unique solutions, the much simpler form of the inco-variance makes it a convenient candidate as a dispersion measure – not only from a computational point of view. As an estimator for $\sigma^2(\mathcal{X})$ we consider the U-statistic

$$\widehat{\sigma}^2(\mathcal{X}) := \frac{1}{m(m-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^m \frac{1}{2} W_r^2(\text{PD}(\mathcal{X}_i), \text{PD}(\mathcal{X}_j)). \quad (1.5)$$

In the following, let $\mathfrak{D} \subseteq \mathbb{R}^d$ be a Borel set. Usually, we choose $\mathfrak{D}$ bounded and convex. Assume that we are given two samples $\mathcal{X}_1, \dots, \mathcal{X}_m$ and $\mathcal{Y}_1, \dots, \mathcal{Y}_n$ stemming from independent stationary, $\mathfrak{D}$ -valued point cloud processes $\mathcal{X} = (\mathcal{X}_t)_t$ and $\mathcal{Y} = (\mathcal{Y}_t)_t$ with common distribution $\mathbb{P}_{\mathcal{X}}$, resp., $\mathbb{P}_{\mathcal{Y}}$. We are interested in testing the second order similarity of corresponding persistence diagrams $\text{PD}(\mathcal{X}_0)$

and $\text{PD}(\mathcal{Y}_0)$ for a given feature dimension $k \in \{0, \dots, d-1\}$ by comparing their Fréchet variances $\text{Var}(\mathcal{X})$ and $\text{Var}(\mathcal{Y})$ or alternatively their independent copy variances $\sigma^2(\mathcal{X})$ and $\sigma^2(\mathcal{Y})$.

Since it is unlikely that these quantities are exactly identical it seems more reasonable to test whether their difference is substantial. This can be formulated as testing problem for relevant differences

$$H_0: (\text{Var}_{\mathcal{X}} - \text{Var}_{\mathcal{Y}})^2 \leq \Delta \quad \text{vs.} \quad H_1: (\text{Var}_{\mathcal{X}} - \text{Var}_{\mathcal{Y}})^2 > \Delta, \quad (1.6)$$

respectively,

$$H_0: (\sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2)^2 \leq \Delta \quad \text{vs.} \quad H_1: (\sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2)^2 > \Delta, \quad (1.7)$$

where $\Delta \geq 0$ is a threshold parameter (set externally).

1.1. A test statistic for relevant differences of Fréchet variances

We begin with the problem (1.6) and test whether the difference $D = \text{Var}(\mathcal{X}) - \text{Var}(\mathcal{Y})$ of the Fréchet variances of the persistence diagrams $\text{PD}(\mathcal{X}_0)$ and $\text{PD}(\mathcal{Y}_0)$ is substantial. To this end, we need to consider the partial-sum process version of the variance estimate, that is

$$\begin{aligned} \widehat{\text{Var}}(\mathcal{X})(s) &= \frac{1}{[ms]} \sum_{i=1}^{\lfloor ms \rfloor} W_r^2(\text{PD}(\mathcal{X}_i), \widehat{\mu}(\mathcal{X})), \quad s \in [0, 1], \\ \widehat{\text{Var}}(\mathcal{Y})(s) &= \frac{1}{[ns]} \sum_{i=1}^{\lfloor ns \rfloor} W_r^2(\text{PD}(\mathcal{Y}_i), \widehat{\mu}(\mathcal{Y})), \quad s \in [0, 1]. \end{aligned}$$

We work with

Assumption 1.1 (r). Let $r > 4$.

- (1) *Moment condition.* $\mathbb{E}[W_r^4(\text{PD}(\mathcal{X}_0), \mu_{\mathcal{X}})] < \infty$, $\mathbb{E}[W_r^4(\text{PD}(\mathcal{Y}_0), \mu_{\mathcal{Y}})] < \infty$.
- (2) *L^r - m -approximability.* The stationary point cloud processes $(\mathcal{X}_t)_t$ and $(\mathcal{Y}_t)_t$ each allow for a Bernoulli shift representation, i.e., $\mathcal{X}_t = f(\varepsilon_t, \varepsilon_{t-1}, \dots)$ and $\mathcal{Y}_t = g(\eta_t, \eta_{t-1}, \dots)$. Here $f: S^\infty \rightarrow \mathfrak{D}$, resp., $g: E^\infty \rightarrow \mathfrak{D}$ are measurable functions and $(\varepsilon_t)_t$, resp., $(\eta_t)_t$ are i.i.d. random elements with values in some measurable space S , resp., E . Furthermore, let $(\mathcal{X}_t^{(m)})_t$ and $(\mathcal{Y}_t^{(m)})_t$ denote the m -dependent approximation processes given by

$$\begin{aligned} \mathcal{X}_t^{(m)} &= f(\varepsilon_t, \dots, \varepsilon_{t-m+1}, \varepsilon_{t-m}^{(t)}, \varepsilon_{t-m-1}^{(t)}, \dots), \\ \mathcal{Y}_t^{(m)} &= g(\eta_t, \dots, \eta_{t-m+1}, \eta_{t-m}^{(t)}, \eta_{t-m-1}^{(t)}, \dots), \end{aligned}$$

where, for each fixed $t \in \mathbb{Z}$, the sequences $(\varepsilon_j^t)_j$, resp., $(\eta_j^t)_j$ are independent copies of the original innovation processes $(\varepsilon_t)_t$, resp., $(\eta_t)_t$. Then $(\mathcal{X}_t)_t$ and $(\mathcal{Y}_t)_t$ are L^r - m -approximable w.r.t. the Hausdorff distance in the sense that

$$\sum_{m \geq 1} \mathbb{E}[d_H(\mathcal{X}_0, \mathcal{X}_0^{(m)})^r]^{1/r} < \infty \quad \text{resp.,} \quad \sum_{m \geq 1} \mathbb{E}[d_H(\mathcal{Y}_0, \mathcal{Y}_0^{(m)})^r]^{1/r} < \infty.$$

- (3) *Finite diagrams.* The number of features in a diagram $\#\text{PD}(\mathcal{X}_0)$, resp., $\#\text{PD}(\mathcal{Y}_0)$ satisfies

$$\mathbb{E}[\#\text{PD}(\mathcal{X}_0)^{4/(r-4)}] < \infty, \quad \text{resp.}, \quad \mathbb{E}[\#\text{PD}(\mathcal{Y}_0)^{4/(r-4)}] < \infty.$$

- (4) *Consistent estimators.* There are unique and consistent estimators $\hat{\mu}(\mathcal{X})$ and $\hat{\mu}(\mathcal{Y})$ for the Fréchet means $\mu(\mathcal{X})$ and $\mu(\mathcal{Y})$ in the sense that

$$W_r(\hat{\mu}(\mathcal{X}), \mu(\mathcal{X})) = o_{\mathbb{P}}(1) \quad (m \rightarrow \infty) \quad \text{resp.}, \quad W_r(\hat{\mu}(\mathcal{Y}), \mu(\mathcal{Y})) = o_{\mathbb{P}}(1) \quad (n \rightarrow \infty).$$

Moreover, these estimators attain the following implicit rate

$$\max_{m^{1/2} \leq k \leq m} \frac{1}{\sqrt{k}} \left| \sum_{i=1}^k W_r^2(\text{PD}(\mathcal{X}_i), \hat{\mu}_{\mathcal{X}}) - W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) \right| = o_{\mathbb{P}}(1) \quad (m \rightarrow \infty)$$

and

$$\max_{n^{1/2} \leq \ell \leq n} \frac{1}{\sqrt{\ell}} \left| \sum_{j=1}^{\ell} W_r^2(\text{PD}(\mathcal{Y}_j), \hat{\mu}_{\mathcal{Y}}) - W_r^2(\text{PD}(\mathcal{Y}_j), \mu_{\mathcal{Y}}) \right| = o_{\mathbb{P}}(1) \quad (n \rightarrow \infty).$$

Moreover, we will rely on the following difference process

$$\hat{D}_{m,n}(s) = s (\widehat{\text{Var}}(\mathcal{X})(s) - \widehat{\text{Var}}(\mathcal{Y})(s)), \quad s \in [0, 1].$$

In particular, we set $\hat{D}_{m,n} = \hat{D}_{m,n}(1) = \widehat{\text{Var}}(\mathcal{X}) - \widehat{\text{Var}}(\mathcal{Y})$. Then, given a tolerance $\Delta > 0$, the null hypothesis in (1.6) is rejected if $\hat{D}_{m,n}^2 > \Delta$. As common in two sample problems, the main difficulty here is to make the statistic $\hat{D}_{m,n}$ pivotal. To that end, we define the self-normalizer

$$\hat{V}_{m,n} = \left\{ \int_0^1 (\hat{D}_{m,n}^2(s) - s^2 \hat{D}_{m,n}^2)^2 \nu(ds) \right\}^{1/2},$$

where ν is some probability measure on $[0, 1]$. Dette and Kokot [14] and Aue et al. [2] study similar normalizers. We have the following result:

Theorem 1.2. *Let the regularity conditions of Assumption 1.1 (r) be satisfied for some $r > 4$ and suppose there is a constant $\tau \in (0, 1)$ such that $\lim_{m,n \rightarrow \infty} m/(m+n) = \tau$. Moreover, let $X_t^* = W_r^2(\text{PD}(\mathcal{X}_t), \hat{\mu}_{\mathcal{X}})$, resp., $Y_t^* = W_r^2(\text{PD}(\mathcal{Y}_t), \hat{\mu}_{\mathcal{Y}})$ for $t \in \mathbb{Z}$. Then*

$$\sqrt{m+n} (\hat{D}_{m,n}^2 - D^2, \hat{V}_{m,n}) \Rightarrow \left(\xi B(1), \xi \left\{ \int_0^1 s^2 (B(s) - sB(1))^2 \nu(ds) \right\}^{1/2} \right),$$

where B is a standard Brownian motion, $\xi = 2\sqrt{\frac{\Gamma_{\mathcal{X}}}{\tau} + \frac{\Gamma_{\mathcal{Y}}}{1-\tau}}(\mathbb{E}[X_0^*] - \mathbb{E}[Y_0^*])$ and where $\Gamma_{\mathcal{X}} = \sum_{k \in \mathbb{Z}} \text{Cov}(X_0^*, X_k^*)$, resp., $\Gamma_{\mathcal{Y}} = \sum_{k \in \mathbb{Z}} \text{Cov}(Y_0^*, Y_k^*)$.